



Brazilian E-Commerce Data Analysis: Full Project Report

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Project: Olist Brazilian E-Commerce Public Dataset Analysis

1. Executive Summary

This project performs a comprehensive end-to-end analysis of the Brazilian E-Commerce Public Dataset by Olist, one of the largest marketplace platforms in Latin America. The dataset contains 100,000+ real e-commerce transactions from 2016 to 2018.

The analysis spans the full data lifecycle: importing raw CSV files, cleaning and transforming data in MySQL, building a star-schema data model in Power BI, and creating an interactive executive dashboard. The project reveals critical insights into revenue trends, customer behavior, logistics performance, and payment preferences, culminating in actionable business recommendations to improve customer retention and operational efficiency.

2. Project Objectives

- Data Quality Assurance:** Clean and transform messy, real-world data to ensure 99%+ data integrity.
- Business Intelligence:** Identify key performance indicators (KPIs) and track business growth over time.
- Customer Behavior Analysis:** Understand purchasing patterns, payment preferences, and geographic distribution.
- Operational Diagnostics:** Investigate the relationship between logistics (delivery time, freight cost) and customer satisfaction (review scores).
- Strategic Recommendations:** Provide data-driven suggestions to reduce cancellations, improve retention, and expand market reach.

3. Methodology

Phase 1: Data Ingestion & Cleaning (MySQL)

- Imported 8 raw CSV files into a relational MySQL database (`olist\_ecommerce\_db`).
- Handled missing values using `NULLIF()` to convert empty strings to proper `NULL`s.
- Converted text-based dates into `DATETIME` format using `ALTER TABLE` and `MODIFY COLUMN`.
- Optimized data types: `DECIMAL(10,2)` for financial columns (price, freight), `INT` for counts, and `VARCHAR` for identifiers.
- Created aggregated tables (e.g., `clean\_payments`) to handle one-to-many relationships.

Phase 2: Exploratory Data Analysis (SQL)

Executed 16 advanced SQL queries covering:

- **Descriptive Analytics:** Total revenue, order distribution, monthly trends.
- **Diagnostic Analytics:** Delivery delays vs. review scores, payment approval times.
- **Customer Analytics:** Retention rates, Customer Lifetime Value (CLV), geographic distribution.
- **Operational Analytics:** Freight cost impact, cancellation patterns, unsold inventory.

Phase 3: Data Modeling & Visualization (Power BI)

- Connected Power BI to MySQL via the MySQL Connector/NET.
- Built a **Star Schema** data model with `olist\_orders\_dataset` as the central fact table.
- Established 1-to-Many relationships with bidirectional cross-filtering for accurate calculations.
- Created DAX measures: `Total Revenue`, `Total Orders`, `Avg Review Score`, `Total Customers`.
- Designed an interactive 4-page dashboard with slicers for State and Time.

4. Key Findings & Business Insights

Financial Performance

- **Total Revenue:** 14 Million BRL from 99,000+ delivered orders.
- **Growth Trend:** Consistent revenue growth from 2016 to 2018, with seasonal spikes in Q4 (Black Friday/Christmas).
- **Top Category:** `beleza\_saude` (Health & Beauty) leads with 1.28M BRL, followed by `relogios\_presentes` (Watches & Gifts).

Customer Behavior

- **Geographic Concentration:** 75%+ of customers are concentrated in Southeast Brazil (São Paulo - SP).
- **Payment Preferences:** Credit Card dominates at 75.32%, followed by Boletão (bank slip) at 19.44%.
- **Retention Rate:** Only ~10-15% of customers are repeat buyers, indicating a heavy reliance on new customer acquisition.

Operational Efficiency

- **Delivery Success Rate:** 97.8% of orders are successfully delivered.
- **Review Distribution:** Classic "J-curve" with 57K five-star reviews, but a concerning cluster of 1-star reviews.
- **Payment Friction:** Average payment approval time is ~10 hours, indicating a bottleneck in the checkout process.
- **Logistics Impact:** Late deliveries directly correlate with a drop in review scores from ~4.3 to ~2.5.

5. Business Recommendations

Based on the data analysis, three strategic recommendations are proposed:

1. Optimize Logistics for Customer Satisfaction

Problem: Late deliveries cause a 40% drop in review scores.

Solution: Partner with regional last-mile delivery startups in the North and Northeast regions. Implement a "Free Shipping Threshold" (e.g., orders over 150 BRL) to offset high freight costs that negatively impact satisfaction.

2. Reduce Boleto Payment Cancellations

Problem: Boleto payments have the highest cancellation rate due to expiration.

Solution: Implement automated WhatsApp/SMS reminders 2 hours before the Boleto expires. This could recover significant lost revenue from the 19.44% of orders using this payment method.

3. Launch a Customer Retention Program

Problem: Only 10-15% of customers make repeat purchases.

Solution: Create an automated email campaign offering a 10% discount to one-time buyers after 90 days. Focus on high-CLV customers identified through the top 10% spenders analysis.

6. Tools & Technologies

| Tool & Purpose |

| **MySQL 8.0** | Database creation, data cleaning, and advanced SQL analysis |

| **Power BI Desktop** | Data modeling, DAX calculations, and interactive dashboard creation |

| **Excel** | Initial data inspection and quick validation |

| **GitHub** | Version control and portfolio hosting |

7. Conclusion

This project demonstrates a complete data analyst workflow: from raw, messy data to actionable business intelligence. The insights reveal that while Olist has strong revenue growth and high delivery success rates, opportunities exist in logistics optimization, payment friction reduction, and customer retention. The interactive Power BI dashboard serves as a decision-support tool for stakeholders to monitor KPIs and drill down into regional performance.

Report prepared by Jwala Kumar | 2026